

Lecture 11: semiconductors, doping, beyond group IV, processing

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MAT SCI 120 — Winter 2026

Last lecture: Connecting Drude's model with quantum mechanical behavior of electrons

Ohm's law: $J = \sigma E$ identifies:

$$\sigma = \frac{N_e e^2 \tau}{m}$$

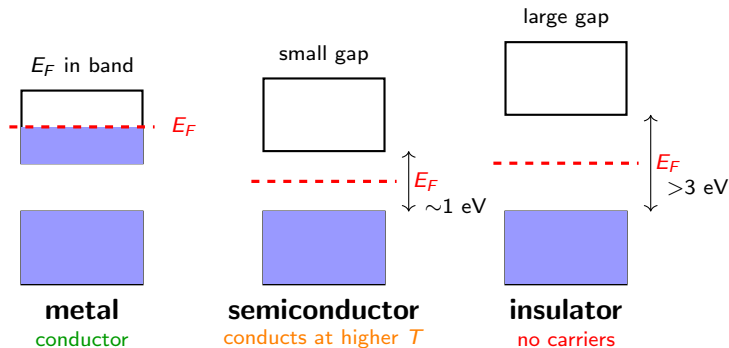
Takeaways:

- ▶ We explained semiconductors or insulators with QM
- ▶ We explained mobility and N_e with effective masses and band occupancy
- ▶ We showed how Fermi-Dirac statistics + density of states explain temperature dependence of conductivity in semiconductor materials
- ▶ We have derived the idea of band structures and electrical properties of materials using quantum mechanics (Kronig-Penney model) and showed they recover the classical results (Drude model).

Main takeaway

Everything relates back to the band structure $E(k)$ of a material!

Remember: materials and their band diagrams



General rule: electrical behavior is explained by mobility and band filling $\rightarrow$ band filling comes from the gap and E_F

Remember: Fermi-Dirac statistics and Boltzmann approximation

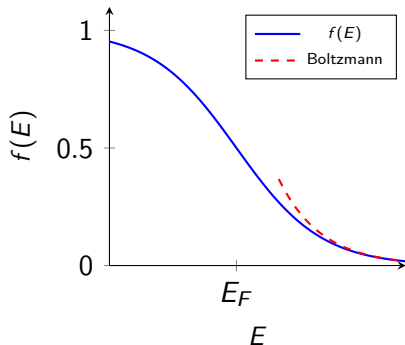
- Occupation probability:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

- When E_F lies in gap far from edges:

$$f(E) \approx \exp\left(-\frac{E - E_F}{k_B T}\right)$$

- Valid for lower temperatures
($E - E_F \gg k_B T$)

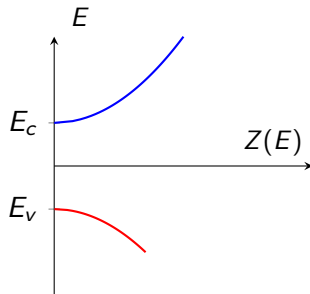


Remember: density of states in 3D parabolic bands

- ▶ Near band edges, $E(\mathbf{k})$ is parabolic
- ▶ DOS counts states per unit energy:

$$Z(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_c}$$

- ▶ Effective mass m^* accounts for band curvature



Remember: Deriving electron concentration in conduction band

- ▶ Integrate DOS $\times$ occupation over conduction band:

$$N_e = \int_{E_c}^{\infty} Z(E) f(E) dE$$

- ▶ Using Boltzmann approximation:

$$N_e = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)$$

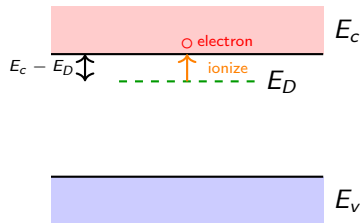
- ▶ N_c is the number of carriers in the conduction band, given by:

$$N_c = 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2}$$

- ▶ $N_c \propto T^{3/2}$ captures thermal broadening of available states

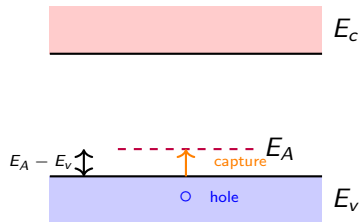
Donor impurities: adding electrons to the conduction band

- ▶ Group V atoms (P, As, Sb) in Si provide extra valence electron
- ▶ Extra electron loosely bound in hydrogen-like orbit
- ▶ Donor level E_D lies just below E_c (typically 10–50 meV)
- ▶ At room T : $k_B T \approx 26$ meV ionizes most donors



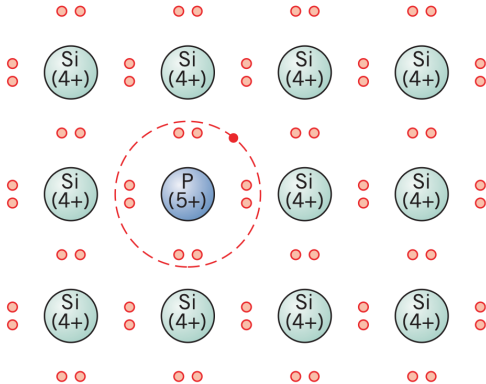
Acceptor impurities: adding holes to the valence band

- ▶ Group III atoms (B, Al, Ga) in Si: one fewer valence electron
- ▶ Acceptor captures electron from valence band $\rightarrow$ mobile hole
- ▶ Acceptor level E_A lies just above E_v (10–50 meV)
- ▶ Ionization: neutral atom $\rightarrow A^- + \text{hole}$

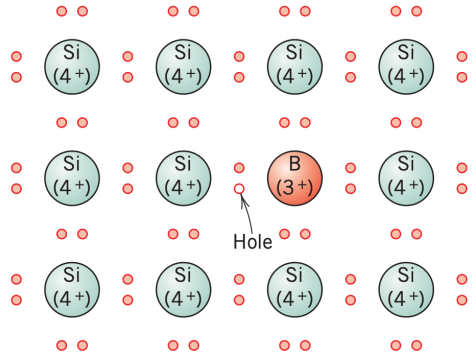


Donor vs acceptor picture

Donor (e.g., P in Si)



Acceptor (e.g., B in Si)



Donor ionization energy: hydrogen-like atom

- ▶ Donor is a hydrogen-like atom in dielectric medium with permittivity ϵ and effective mass m^*
- ▶ Bohr model scaled: binding energy reduced by factor $(m^*/m_0)(1/\epsilon^2)$

$$E_D = \frac{m_e^*}{m_0} \cdot \frac{1}{\epsilon^2} \cdot 13.6 \text{ eV}$$

- ▶ **Example for Si:** $\epsilon \approx 12$, $m^* \approx 0.3 m_0$

$$E_D \approx \frac{0.3}{144} \times 13.6 \text{ eV} \approx 30 \text{ meV}$$

- ▶ Compare to hydrogen: 13.6 eV $\rightarrow$ shallow donors easily ionized

Hydrogen-like atom: radius

- ▶ Donor is a hydrogen-like atom in dielectric medium with permittivity ε and effective mass m^*
- ▶ Radius from Bohr model: $a_0 = \frac{4\pi\varepsilon_0\hbar^2}{e^2 m_e}$
- ▶ **Example for Si:** $\varepsilon \approx 12$, $m^* \approx 0.3 m_0$
- ▶ What is the radius of the electron for the donor?

Charge neutrality condition

- ▶ Crystal must remain electrically neutral
- ▶ **Positive charges:** holes (N_h) and ionized donors (N_D^+)
- ▶ **Negative charges:** electrons (N_e) and ionized acceptors (N_A^-)

$$N_e + N_A^- = N_h + N_D^+$$

- ▶ This constraint determines Fermi level position for any doping profile
- ▶ Combined with mass action law $N_e N_h = N_i^2$, enables calculation of all carrier densities

$$\begin{array}{ll} \text{n-type:} & N_D^+ \gg N_A^- \Rightarrow N_e \approx N_D^+ \\ \text{p-type:} & N_A^- \gg N_D^+ \Rightarrow N_h \approx N_A^- \end{array}$$

Mass action law is a thermodynamic equilibrium constraint

Notation: N_e = electron concentration, N_h = hole concentration, N_i = intrinsic carrier concentration

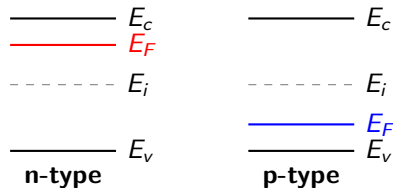
- ▶ Product of electron and hole concentrations fixed at given temperature
- ▶ Holds regardless of doping level—adding donors increases N_e , decreases N_h
- ▶ Reflects dynamic equilibrium: thermal generation rate = recombination rate
- ▶ Combined with charge neutrality, enables solving for N_e , N_h , and E_F

Mass Action Law

$$N_e N_h = N_i^2 \quad (1)$$

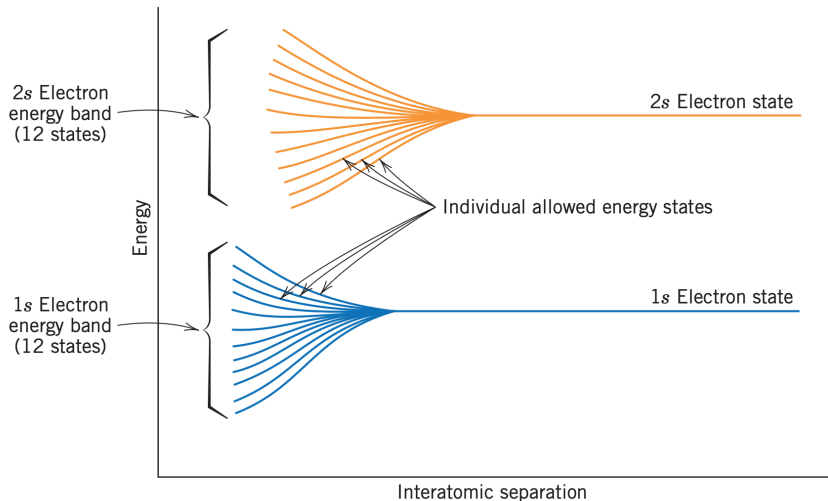
Fermi level shift with doping

- ▶ **n-type** ($N_D \gg N_A$): E_F moves toward E_c ;
 $N_e \approx N_D$
- ▶ **p-type** ($N_A \gg N_D$): E_F moves toward E_v ;
 $N_h \approx N_A$
- ▶ Higher doping $\rightarrow E_F$ closer to band edge
- ▶ Temperature affects ionization fraction



Conceptual questions

Q: Using what you learned in MAT SCI 120, explain the following diagram. What does

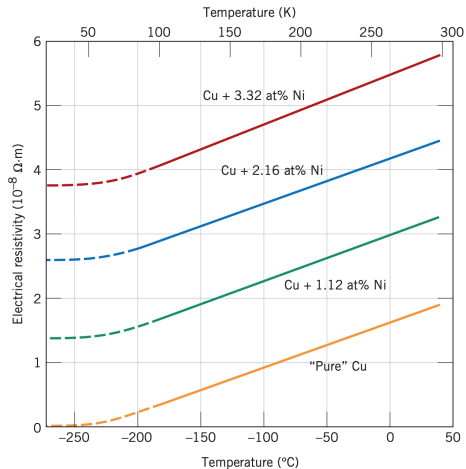


it mean?

Conceptual questions

Q: Using what you learned in MAT SCI 120, explain the diagram on the right.

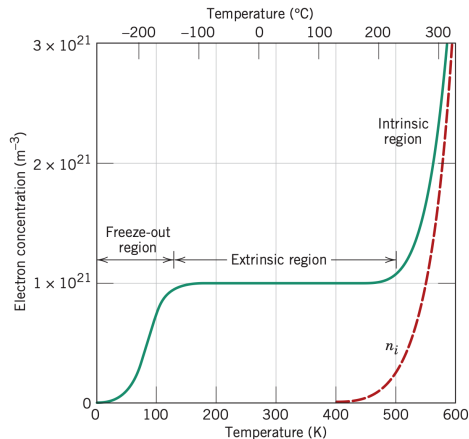
- ▶ What is the *physical meaning* of each regime/feature?
- ▶ What *mechanism* explains the trend(s)?
- ▶ Use the Drude model ($\sigma = N_e e^2 \tau / m$) and/or the wave picture (band structure etc.) to justify your explanation.
- ▶ Identify which parameter(s) change most with temperature for this material: N_e , τ , m^* and why.



Conceptual questions

Q: Using what you learned in MAT SCI 120, explain the diagram on the right.

- ▶ What is the *physical meaning* of each regime/feature?
- ▶ What *mechanism* explains the trend(s)?



Conceptual questions

1. Two semiconductors have identical τ and identical donor density, but different m^* . Which has higher σ and justify using both *mobility* and *DOS* arguments?
2. Why adding donors as dopants increases N_e but *reduces* N_h (even though donors do not directly “remove holes”)?
3. The 3D DOS near a band edge scales as $Z(E) \propto (m^*)^{3/2} \sqrt{E - E_c}$. If m^* increases, what happens to:
 - ▶ the number of available states near E_c ,
 - ▶ mobility (qualitatively),
 - ▶ and the net conductivity at fixed E_F ?

Summary so far

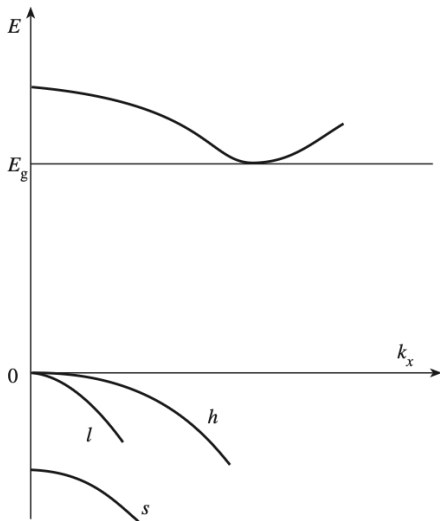
- ▶ We explained that electrons and holes have an effective mass, allowing us to simplify our formalism.
- ▶ Effective masses allow us to simplify the problem to a “free electron” moving with an effective mass, thus producing a mobility.
- ▶ We computed N_c , N_v as a function of the temperature, gap, and the dopant concentration.

Main goal

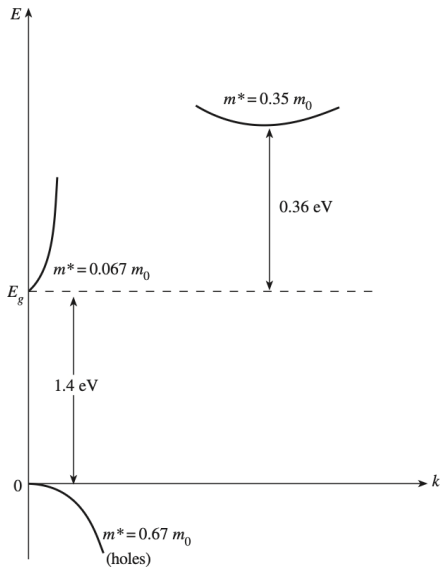
What additional characteristics should we know to engineer devices?

Real band structures (simple picture)

Si



GaAs

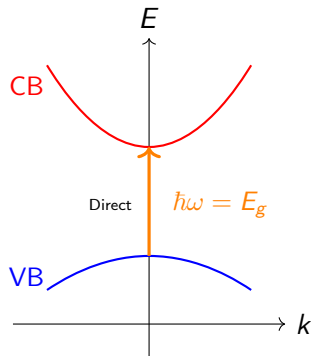


Direct band gap semiconductors

- ▶ CB minimum and VB maximum at **same** k
- ▶ Photon momentum $p = \hbar\omega/c$ is **negligible** compared to electron crystal momentum $\hbar k$
- ▶ Optical transitions are **vertical** in $E(k)$ diagrams ($\Delta k \approx 0$)

Electron excitation by a photon with a direct band gap material:

$$\hbar\omega = E_g$$



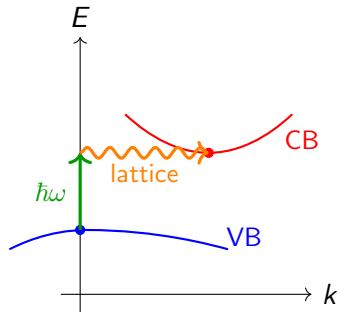
Indirect band gap semiconductors

Electron excitation by a photon + lattice with an indirect band gap material:

$$\hbar\omega_{\text{photon}} \approx E_g$$

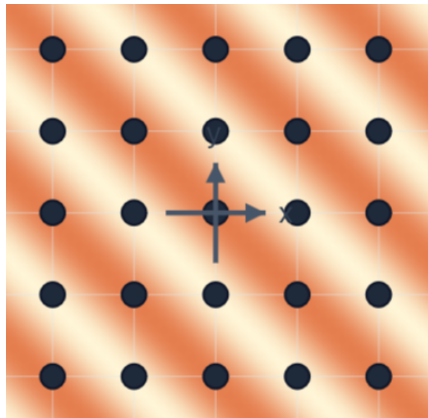
$$k_{\text{lattice}} \approx \Delta k$$

- ▶ Second-order process → **lower probability** than direct absorption
- ▶ k_{lattice} is similar to a vibration in the lattice



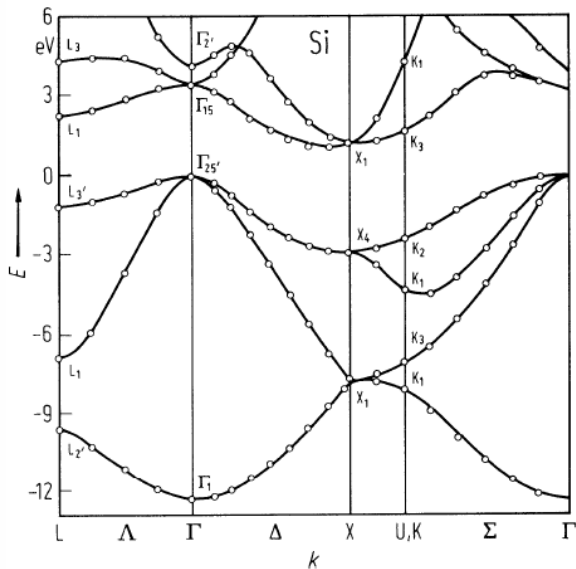
Why real band structures depend so much on k ? Bragg diffraction

- ▶ Remember that diffraction happens whenever the difference between the incoming wavevector k and the outgoing wavevector k' is a reciprocal lattice vector G
- ▶ This means that electrons also undergo “Bragg diffraction” inside the crystal according to the direction
- ▶ See demo: “direction” (k) of the electron creates different standing waves



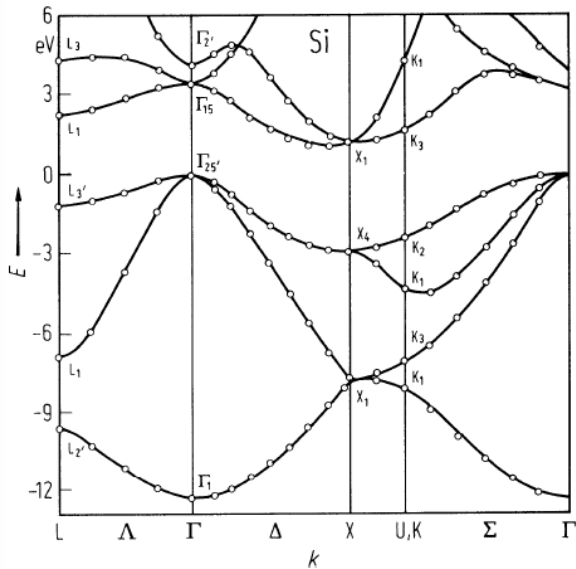
Band structure of real semiconductors: Si

- ▶ Do not panic! This is just an energy vs k plot like before.
- ▶ The problem is that $\mathbf{k}$ is a 3D vector now. To plot a 1D graph, we have to select a few important directions (the Greek letters)
- ▶ We will identify: the gap (indirect gap), the valence band, and the conduction band.



Important points to remember on real band structures

- ▶ Often, the energy 0 is set to E_F (unless said otherwise).
- ▶ If you see a gap, this is a semiconductor or insulator. If E_F crosses bands, this is likely metallic.
- ▶ One reason why you see many bands is because there are many orbitals, e.g., 3s and 3p in this silicon case.
- ▶ Do not worry about the k -axis at this point. All you need to remember is that they indicate “directions” in the reciprocal space



Comparing semiconductors

Table: Energy gaps, effective masses, and relative dielectric constants (ϵ_r) for selected semiconductors (at 273 K).

Semiconductor	E_g (eV)	Effective mass m^*/m		ϵ_r
		Electrons	Holes	
Ge	0.67	0.200	0.3	16
Si	1.14	0.330	0.5	12
InSb	0.16	0.013	0.6	18
InAs	0.33	0.020	0.4	14.5
InP	1.29	0.070	0.4	14
GaSb	0.67	0.047	0.5	15
GaAs	1.39	0.072	0.5	13

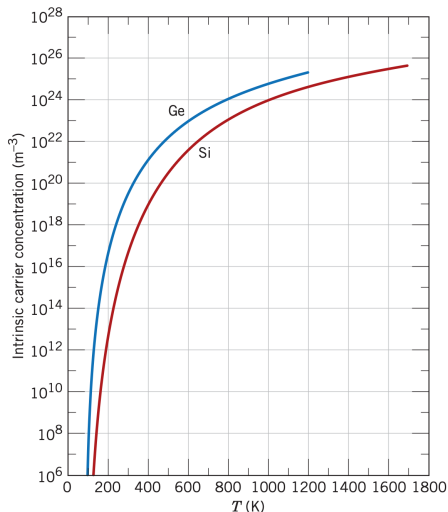
Why does the gap changes so much with composition?

- ▶ **Band gap origin:** in a tetrahedrally bonded solid, the valence band is mostly *bonding* states and the conduction band mostly *antibonding* states. The gap scale is controlled by the barrier height and width in the K-P model.
- ▶ **GaAs (III–V):** Ga–As bonds are largely covalent (sp^3 network) *but* are more **ionic** (electronegativity difference) than Si. This creates a deeper well. Orbital differences make it such that the VBM is more As-like and the CBM more Ga-like.
- ▶ Why is that? Think about what happens in ionic-like bonds: electrons occupy states of the more electronegative element (i.e., valence band), and the outer shell of the less electronegative element becomes unoccupied (i.e., conduction band).
- ▶ Result: a higher band gap (~ 1.4 eV near room temperature) compared to silicon.

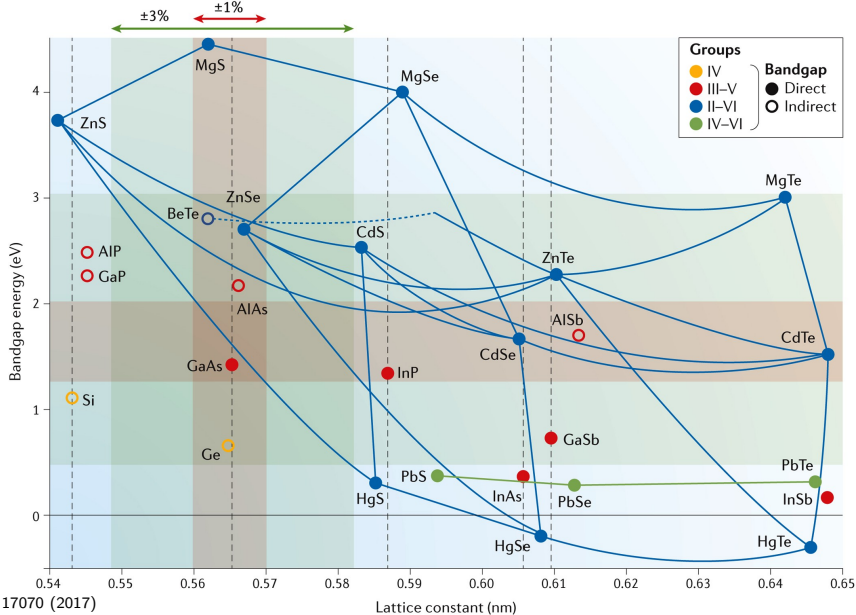
Conceptual questions

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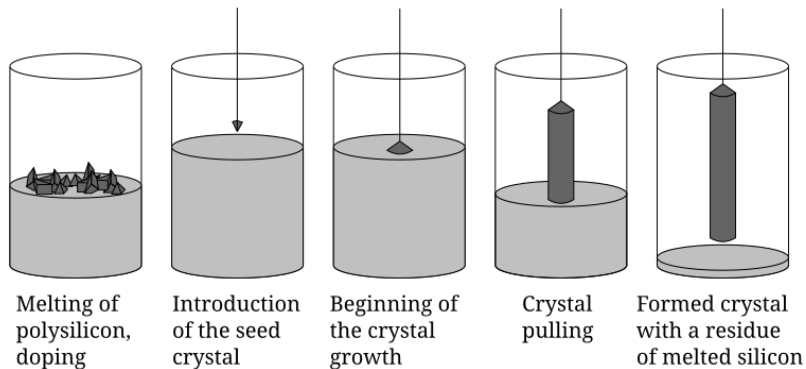
- ▶ Why is the curve behaving like it is?
- ▶ What explains the trends?
- ▶ Why the concentration for Ge is higher than Si?
- ▶ How would this curve look like for GaAs?



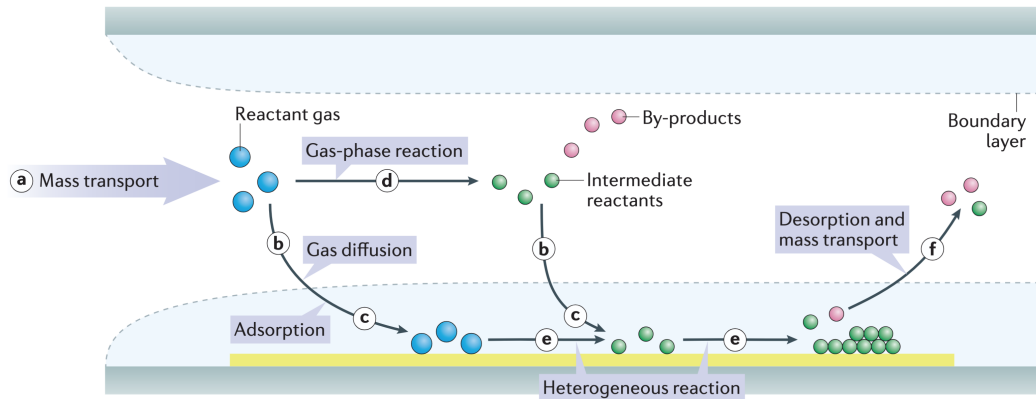
Engineering semiconductors: creating semiconductor alloys



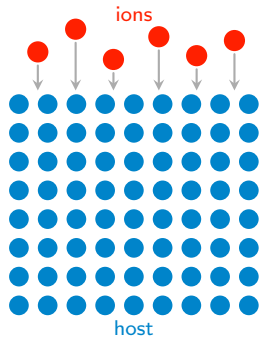
Making silicon: Czochralski process



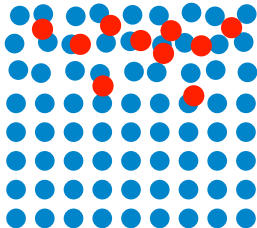
Making semiconductors: chemical vapor deposition



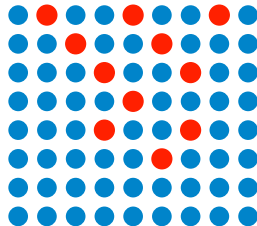
Doping semiconductors: ion implantation



implantation process



after implantation



after annealing